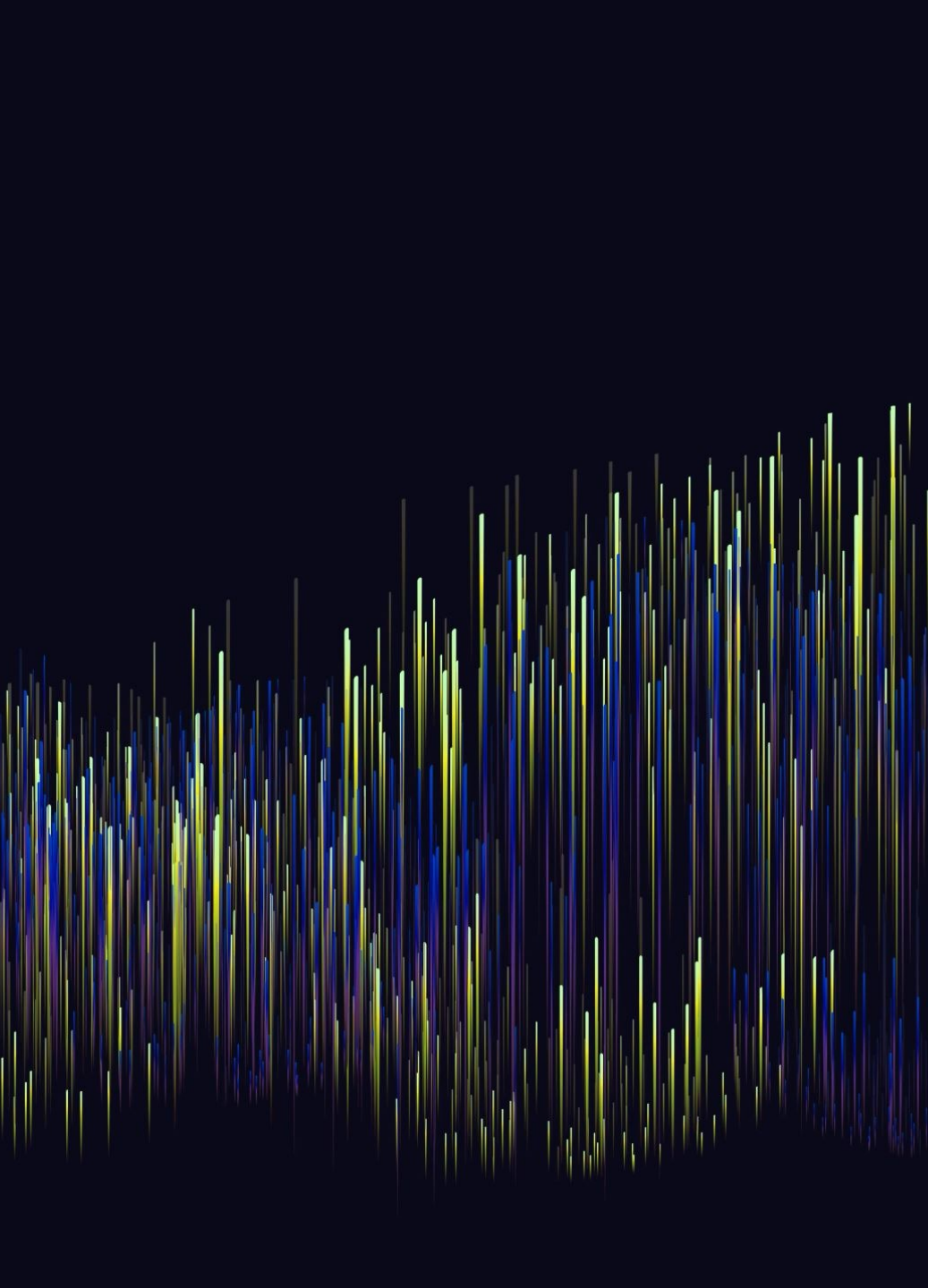


A decorative background on the left side of the slide, featuring a dark blue field with numerous vertical lines of varying heights and colors, including yellow, light blue, and white. A vertical red bar is visible on the far left edge.

Stat 3113: Lesson 12

The Binomial Probability
Distribution

Binomial Random Variable

➤ Conditions:

1. n independent, identical trials.
2. Each trial results in either a Success or a Failure.
3. $P(\text{success}) = p$ and $P(\text{failure}) = 1 - p = q$, i.e., these values are constant from trial to trial.
4. Let $X = \# \text{ of successes in } n \text{ trials}$.

Then, X is a **Binomial random variable**.

➤ Special Notation: $X \sim \text{Bin}(n, p)$

Ex: $X \sim \text{Bin}(20, 0.15)$ means X is a binomial rv with $n = 20$ and $p = 0.15$.

Binomial Distribution: pmf and cdf.

- pmf of Binomial rv with parameters n and p : $\rightarrow$ "n choose x" = $\frac{n!}{x!(n-x)!}$

$$f(x) = P(X = x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

- cdf of Binomial rv with parameters n and p :

$$F(x) = P(X \leq x) = \sum_{y=0}^x b(y; n, p) = \sum_{y=0}^x \binom{n}{y} p^y (1-p)^{n-y} \quad x = 0, 1, \dots, n$$

- In Excel:

- pmf: $P(X = x) \rightarrow = \text{binom.dist}(x, n, p, \text{false})$

- cdf: $P(X \leq x) \rightarrow = \text{binom.dist}(x, n, p, \text{true})$

Example 1: $X \sim \text{Bin}(20, 0.15)$ Find the following:

a. $P(X = 8) = \binom{20}{8} (0.15)^8 (1 - 0.15)^{(20-8)} \approx \text{binom.dist}(8, 20, 0.15, \text{false}) \approx 0.0046$

$$\begin{aligned}
 \text{b. } P(X \leq 2) &= F(2) \\
 &= \sum_{y=0}^2 \binom{20}{y} (0.15)^y (1-0.15)^{20-y} \\
 &= \text{binom.dist}(2, 20, 0.15, \text{true}) \approx 0.4049
 \end{aligned}$$

$$\begin{aligned}
 \text{c. } P(X \geq 1) &= 1 - P(X < 1) = 1 - P(X \leq 0) = 1 - F(0) \\
 &= 1 - \text{binom.dist}(0, 20, 0.15, \text{true}) \approx 0.9612.
 \end{aligned}$$

Expected Value, Variance, and Standard Deviation of Binomial RV

$$\begin{array}{lll}
 1. E(X) = np & 2. V(X) = np(1-p) & 3. \sigma_X = \sqrt{V(X)}
 \end{array}$$

$\downarrow$
Proof: 4 ways to prove it!
 + 2 bonus points $\rightarrow$ Exam!

Example 2: $X \sim \text{Bin}(20, 0.15)$ Find the following:

a. $E(X) = np = 20 \times 0.15 = 3.$

b. $V(X) = np(1-p) = 20 \times 0.15 \times (1-0.15) = 2.55$

c. $\sigma_X = \sqrt{V(X)} \simeq 1.5969.$

Example 3: A company producing fine crystal knows from experience that 10% of its goblets have cosmetic flaws and must be classified as “seconds”. A random sample of six goblets is taken.

a. What is the probability exactly one is a “second”?

$$X = \# \text{ seconds selected.} \Rightarrow X \sim \text{Bin}(6, 0.1)$$

$$P(X=1) = \text{binom.dist}(1, 6, 0.1, \text{false}) \simeq 0.3543. \simeq 35\%$$

b. What is the probability of at least two “seconds”?

$$\begin{aligned} P(X \geq 2) &= 1 - P(X < 2) \\ &= 1 - P(X \leq 1) \\ &= 1 - F(1) = 1 - \text{binom.dist}(1, 6, 0.1, \text{true}) \\ &= 0.1143. \end{aligned}$$

c. What is the average number of “seconds” in the sample of six?

$$E[X] = np = 6 \times 0.10 = 0.6$$

On average, we should expect having 1 second in a sample of six goblets.

NOTE: A Binomial Random Variable Is a Sum of Bernoulli Random Variables.

Estimating the Success Probability

- For $X \sim \text{Ber}(p)$, the success probability p is usually unknown.
- If n independent Bernoulli trials are conducted and the number of successes is counted. To estimate the success probability p , we compute the sample proportion $\hat{p}$.

$$\hat{p} = \frac{\text{number of successes}}{\text{number of trials}} = \frac{X}{n}$$

↑
Relative Frequency!

- Since $\hat{p}$ is a statistic, its value can change from sample to sample, i.e. there is uncertainty in $\hat{p}$.
- This estimate is unbiased (Its expected value exactly equals to the quantity it is estimating). The success probability, which is unknown, is denoted by p . The sample proportion, which is known, is denoted by $\hat{p}$. The “hat” (^) indicates that $\hat{p}$ is used to estimate the unknown value p .
- **Uncertainty of $\hat{p}$:**

$$\sigma_{\hat{p}} = \sqrt{\text{Var}(\hat{p})} = \sqrt{\frac{p(1-p)}{n}} \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \hat{\sigma}_{\hat{p}}$$

Example 4: A quality engineer is testing the calibration of a machine that packs ice cream into containers. In a sample of 20 containers, 3 are underfilled. Estimate the probability p that the machine underfills a container.

$X = \#$ underfilled containers.

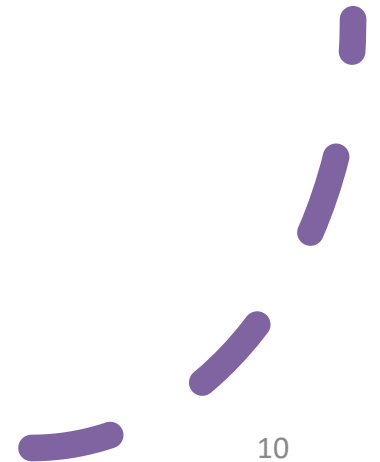
$$\Rightarrow \hat{p} = \frac{X}{n} = \frac{3}{20} = 0.15$$

$$\Rightarrow \hat{\sigma}_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.15(1-0.15)}{20}} \approx 0.0794.$$

The end!

Reading

Chapter 4 – Sections 4.1, 4.2



Next Topic



The **Poisson** Probability Distribution



Chapter 4 - Sections 4.1, 4.2